Order of Operations with Radicals

Question Bank 4.06

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $20 - 3 \cdot 4 + 2$

2. $24 \div 4 \cdot 3$

3. $5 + 2(7 - 3)^2$

4. $\frac{4^2 + 8}{2^3 - 5}$

5. $-3^2 + (-3)^2$

6. $\sqrt{100 - 36} + 2^{-2}$

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $4(8.63-(3.2)^2)+\sqrt[3]{125}$

b. $\frac{2^3-5}{6^2}-\sqrt{\frac{4}{9}-\frac{1}{3}}$

c. $12\left(\left(\frac{1}{2}\right)^3+3^{-2}\right)$

8. Evaluate each expression.

a. $[3+(2-5)^2]\cdot 2^3$

b. $\frac{(-4)^3}{2^4}-\sqrt{81}$

c. $\sqrt[3]{-27}\cdot\left(\frac{1}{3}\right)^{-2}$

Working backwards.

9. Find the value of n that makes each statement true.

a. $n + 3 \cdot 4 = 20$

b. $(n+1)^2 = 49$, where n is positive

c. $\frac{n-4}{2^2} = 5$

Reading someone else's work.

10. Deshawn says that $12 \div 3 \cdot 2 = 2$, because you always multiply before you divide. Is he right? Explain, and give the correct value.

11. Which is the value of $18 - 2(3)^2$?

- ○ A. 12
- ○ B. 144
- ○ C. −18
- ○ D. 0

12. Select ALL of the expressions whose value is 4.

- ☐ A. -2^2
- ☐ B. $\left(\frac{1}{2}\right)^{-2}$
- ☐ C. $2+3-1$
- ☐ D. $\sqrt{16}$
- ☐ E. $\frac{6^2-4}{2^3}$
- ☐ F. $12 \div 3 \cdot 2$